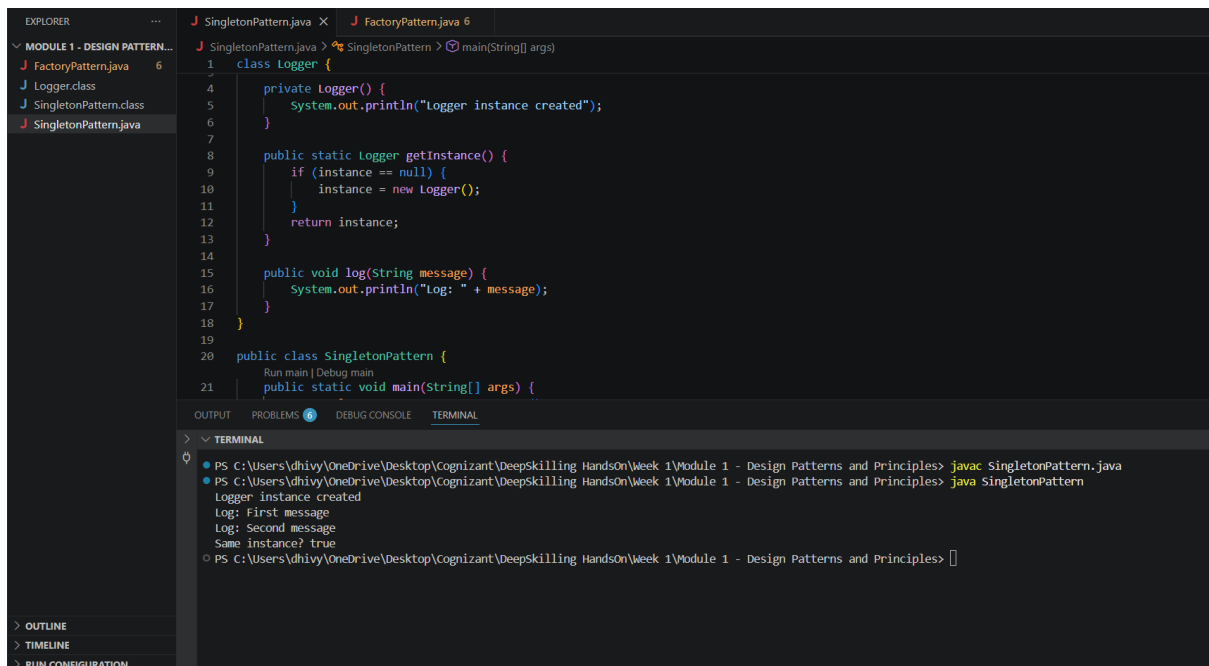


# CTS DIGITAL NURTURE 5.0 – DEEPSKILLING OUTPUTS

## Module 1 - Design Patterns and Principles

### Exercise 1: Implementing the Singleton Pattern

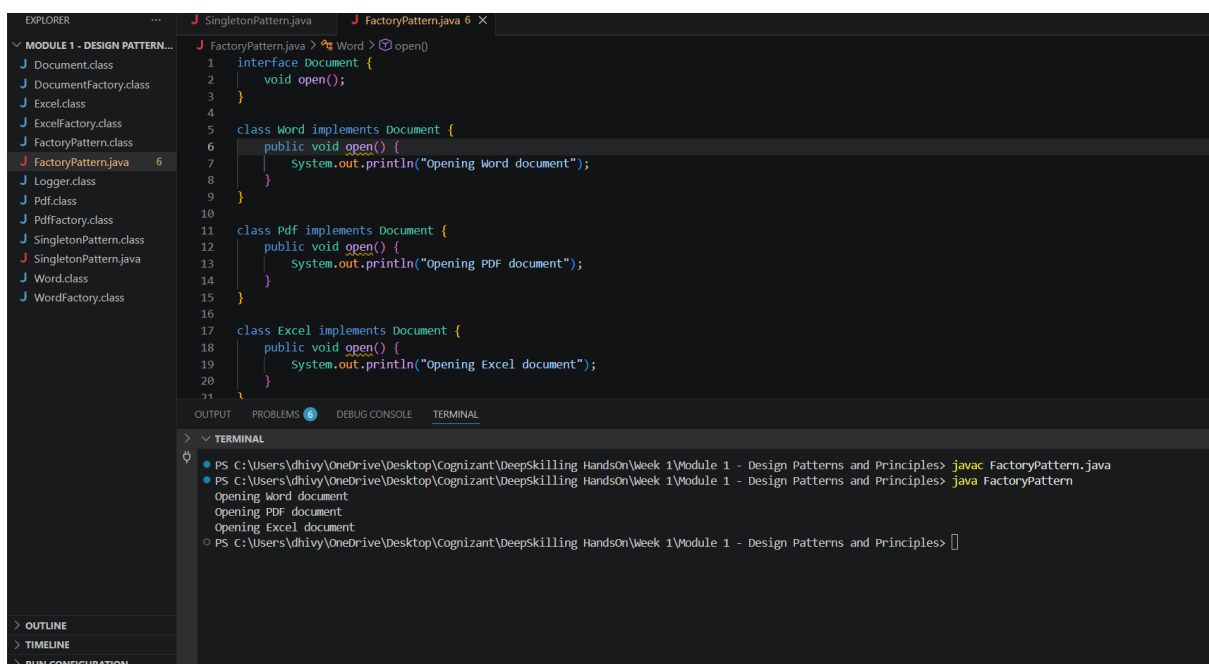


```
1 class Logger {
2
3
4     private Logger() {
5         System.out.println("Logger instance created");
6     }
7
8     public static Logger getInstance() {
9         if (instance == null) {
10             instance = new Logger();
11         }
12         return instance;
13     }
14
15     public void log(String message) {
16         System.out.println("Log: " + message);
17     }
18 }
19
20 public class SingletonPattern {
21     Run main | Debug main
22     public static void main(String[] args) {
```

OUTPUT PROBLEMS DEBUG CONSOLE TERMINAL

PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 1 - Design Patterns and Principles> javac SingletonPattern.java  
PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 1 - Design Patterns and Principles> java SingletonPattern  
Logger instance created  
Log: First message  
Log: Second message  
Same instance? true  
PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 1 - Design Patterns and Principles>

### Exercise 2: Implementing the Factory Method Pattern



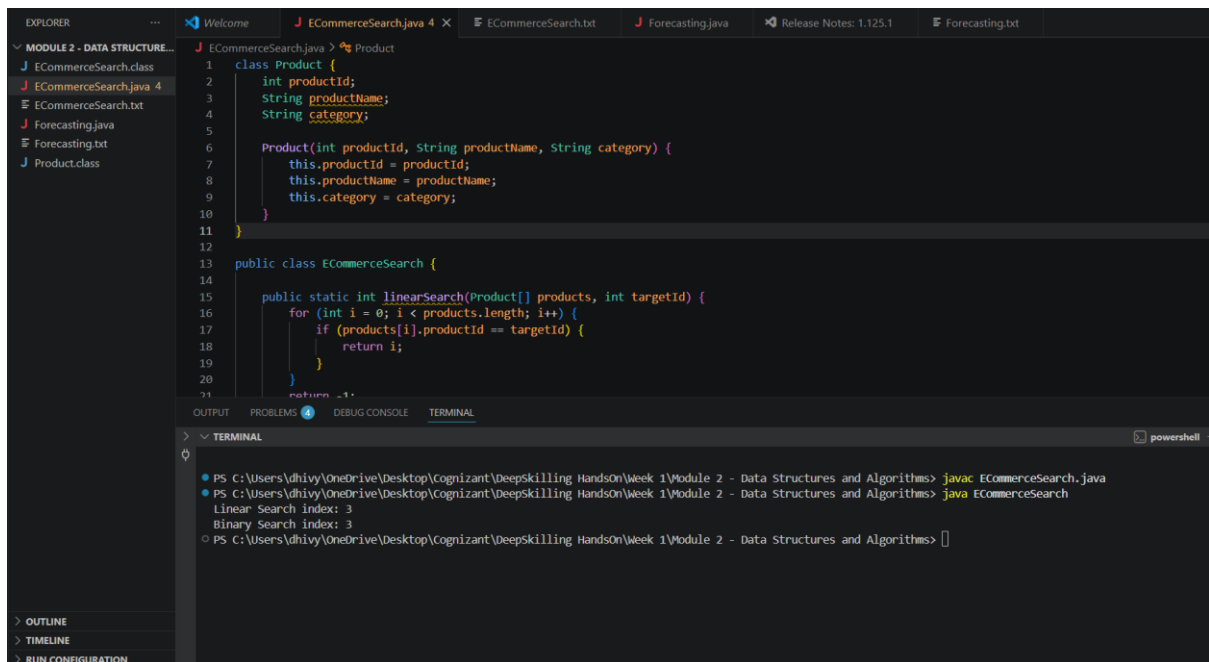
```
1 interface Document {
2     void open();
3 }
4
5 class Word implements Document {
6     public void open() {
7         System.out.println("Opening Word document");
8     }
9 }
10
11 class Pdf implements Document {
12     public void open() {
13         System.out.println("Opening PDF document");
14     }
15 }
16
17 class Excel implements Document {
18     public void open() {
19         System.out.println("Opening Excel document");
20     }
21 }
```

OUTPUT PROBLEMS DEBUG CONSOLE TERMINAL

PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 1 - Design Patterns and Principles> javac FactoryPattern.java  
PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 1 - Design Patterns and Principles> java FactoryPattern  
Opening Word document  
Opening PDF document  
Opening Excel document  
PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 1 - Design Patterns and Principles>

## Module 2 - Data Structures and Algorithms

### Exercise 2: E-commerce Platform Search Function



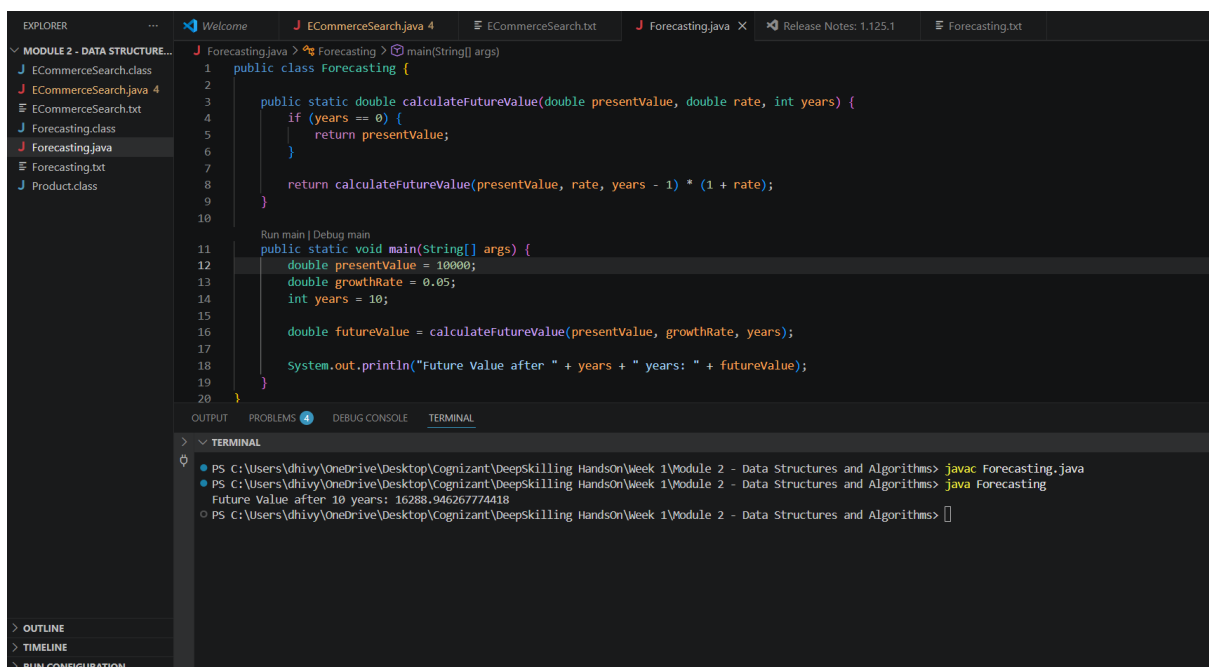
```
1 class Product {
2     int productId;
3     String productName;
4     String category;
5
6     Product(int productId, String productName, String category) {
7         this.productId = productId;
8         this.productName = productName;
9         this.category = category;
10    }
11 }
12
13 public class ECommerceSearch {
14
15     public static int linearSearch(Product[] products, int targetId) {
16         for (int i = 0; i < products.length; i++) {
17             if (products[i].productId == targetId) {
18                 return i;
19             }
20         }
21         return -1;
22     }
23 }
```

OUTPUT PROBLEMS DEBUG CONSOLE TERMINAL

TERMINAL

- PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 2 - Data Structures and Algorithms> javac ECommerceSearch.java
- PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 2 - Data Structures and Algorithms> java ECommerceSearch
- Linear Search index: 3
- Binary Search index: 3
- PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 2 - Data Structures and Algorithms> []

### Exercise 7: Financial Forecasting



```
1 public class Forecasting {
2
3     public static double calculateFutureValue(double presentValue, double rate, int years) {
4         if (years == 0) {
5             return presentValue;
6         }
7
8         return calculateFutureValue(presentValue, rate, years - 1) * (1 + rate);
9     }
10 }
11
12 Run main | Debug main
13 public static void main(String[] args) {
14     double presentValue = 10000;
15     double growthRate = 0.05;
16     int years = 10;
17
18     double futureValue = calculateFutureValue(presentValue, growthRate, years);
19     System.out.println("Future Value after " + years + " years: " + futureValue);
20 }
```

OUTPUT PROBLEMS DEBUG CONSOLE TERMINAL

TERMINAL

- PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 2 - Data Structures and Algorithms> javac Forecasting.java
- PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 2 - Data Structures and Algorithms> java Forecasting
- Future Value after 10 years: 16288.94626774418
- PS C:\Users\dhivy\OneDrive\Desktop\Cognizant\DeepSkilling HandsOn\Week 1\Module 2 - Data Structures and Algorithms> []